Introduction; Circuit Elements; Ohm's Law; KCL

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August 18, 2026

Basic Electrical Quantities

- Basic quantities: current, voltage and power
  - *Current*: time rate of change of electric charge

    $$I = dq/dt$$

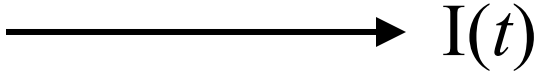
    1 Amp = 1 Coulomb/sec
  - *Voltage*: electromotive force or potential, V

    1 Volt = 1 Joule/Coulomb = 1 N·m/coulomb
  - *Power*: $P = I\,V$

    1 Watt = 1 Volt·Amp = 1 Joule/sec

Current, I

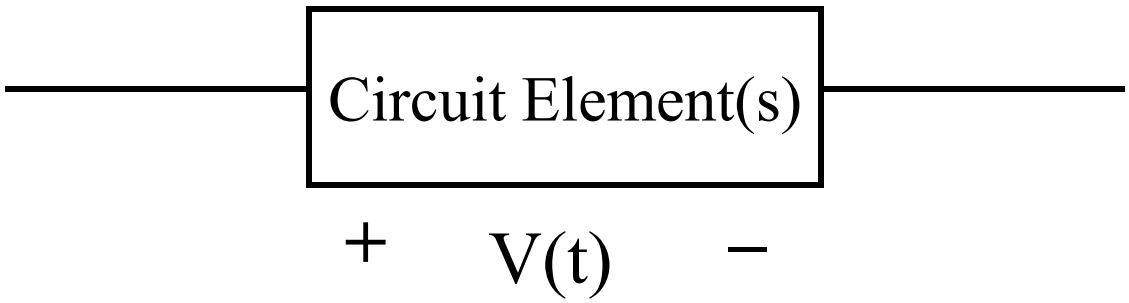
- Normally we talk about the movement of positive charges although we know that, in general, in metallic conductors current results from electron motion (conventionally positive flow)
- The sign of the current indicates the direction of flow



- Types of current:
  - direct current (dc): batteries and some special generators
  - alternating current (ac): household current which varies with time

Voltage, V

Voltage is the difference in energy level of a unit charge located at each of two points in a circuit, and therefore, represents the energy required to move the unit charge from one point to the other



Default Sign Convention

- *Passive sign convention* : current should enter the positive voltage terminal



- Consequence for $P = I\,V$
  - Positive (+) Power: element absorbs power
  - Negative (-) Power: element supplies power

Electrical Analogies (Physical)

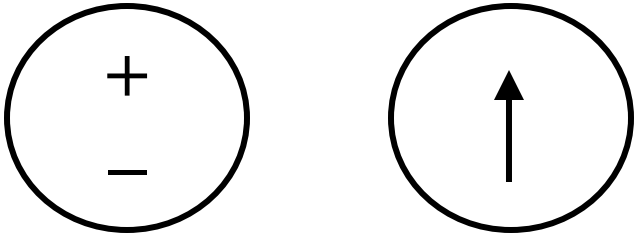
	Electrical	Hydraulic
Base quantity	Charge (q)	Mass (m)
Flow variable	Current (I)	Fluid flow (G)
Potential variable	Voltage (V)	Pressure (p)

Active vs. Passive Elements

- *Active elements* can generate energy
  - Voltage and current sources
  - Batteries
- *Passive elements* cannot generate energy
  - Resistors
  - Capacitors and Inductors (but CAN store energy)

Independent Sources

An *independent source* (voltage or current) may be DC (constant) or time-varying (AC), but does not depend on other voltages or currents in the circuit



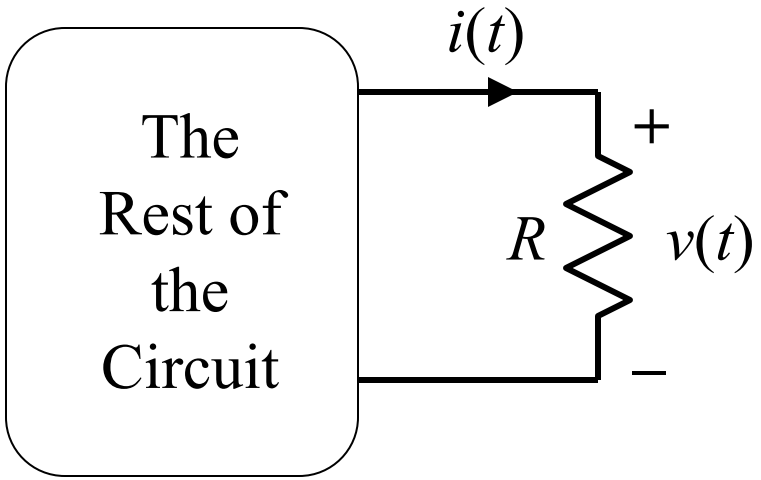
Resistors

- A resistor is a circuit element that dissipates electrical energy (usually as heat)
- Real-world devices that are modeled by resistors: incandescent light bulbs, heating elements (stoves, heaters, etc.), long wires
- Resistance is measured in Ohms (Ω)

Ohm's Law

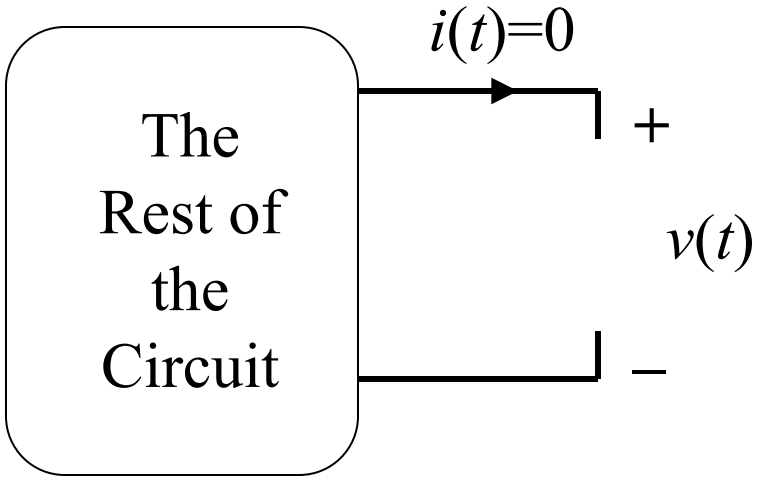
$v(t) = i(t)\,R$ - or - $V = I\,R$

$p(t) = i^2(t)\,R = v^2(t)/R$ [+ (absorbing)]



Open Circuit

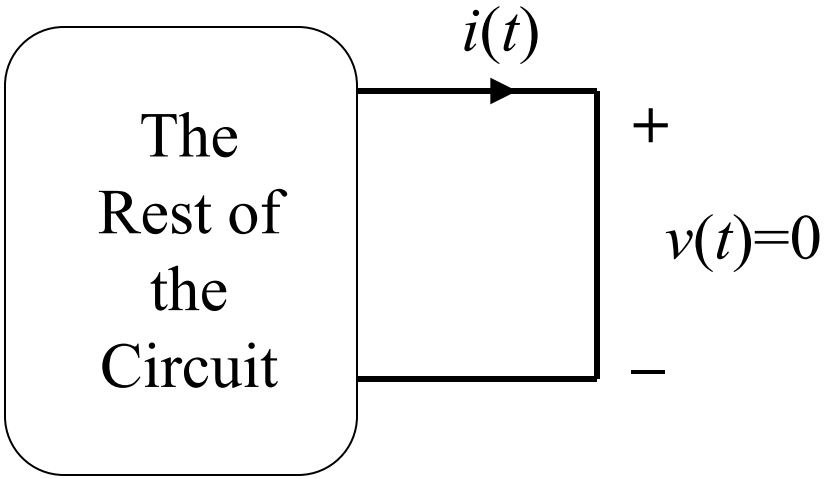
- What if R=∞?



- i(t) = v(t)/R = 0

Short Circuit

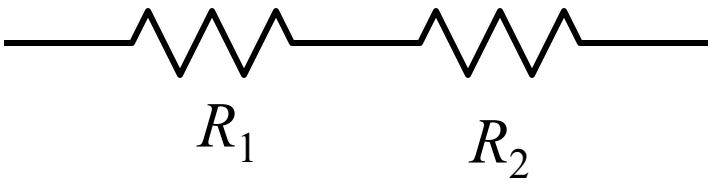
- What if R=0?



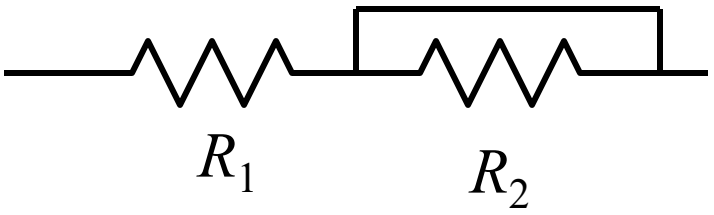
- v(t) = R i(t) = 0

Series

Two elements are in *series* if the current that flows through one must also flow through the other.



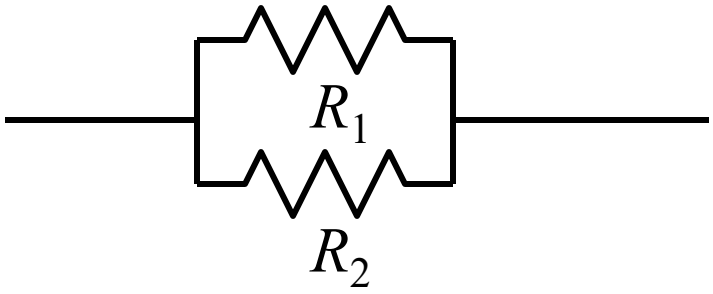
Series



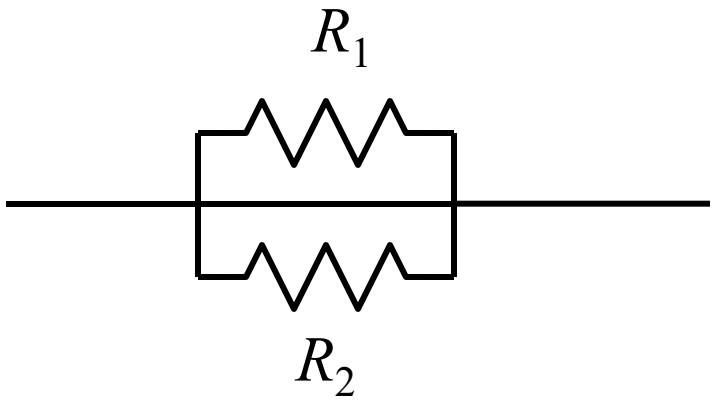
Not Series

Parallel

Two elements are in parallel if they are connected between (share) the same two (distinct) end nodes.



Parallel

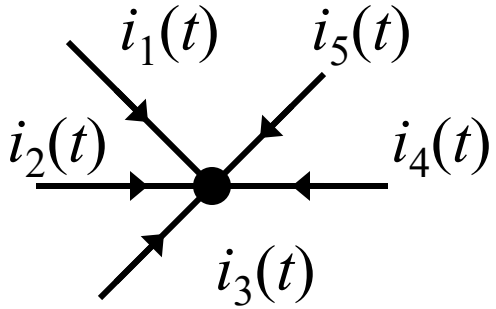


Not Parallel

Kirchhoff's Laws

- Kirchhoff's Current Law (KCL)
  - sum of all currents entering a node is zero
  - sum of currents entering node is equal to sum of currents leaving node
- Kirchhoff's Voltage Law (KVL)
  - sum of voltages around any loop in a circuit is zero

KCL (Kirchhoff's Current Law)



The sum of currents entering the node is zero:

$$\sum_{j=1}^{n} i_j(t) = 0$$

Analogy: mass flow at pipe junction

Class Examples

- Drill Problems 1, 2, 4